Oracle-Blind Rollout Geometry Cannot Distinguish Reward Hacking from Benign Convergence

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Abstract

A reward hack and a benign capability improvement are, at the level of the rollout distribution, the *same event*: a policy sharpening onto whatever the reward favors. The only quantity that distinguishes them—whether the favored mode also has higher *true quality*—is the held-out oracle, which an *oracle-blind* detector cannot observe by construction, so no oracle-blind geometric statistic can separate reward hacking from benign convergence *by kind*; we formalize this as Proposition 1. We support the claim with three empirical legs. First, in a detector-characterization study, every candidate geometric axis (convergence sharpness, output degeneracy, advantage-distribution shape) fires *identically* on a benign control matched on that axis—including `return 0` (a degenerate hack) versus `return n*(n+1)//2` (a correct one-liner), which are geometrically indistinguishable. Second, two independent real benign GRPO runs (base Qwen2.5-1.5B, a well-specified coding task on which no hacking is possible) each produced a **false positive**—a gentle benign convergence at KL ≈ 0.02 latched a detector onset at step 94 at a warmup-normalized CUSUM of $S \approx 37.5$, roughly eight times the firing threshold, reproduced across seeds—and an onset-strength sweep establishes a **crossover** below which any onset is inseparable from benign convergence for *any* threshold. Third, an elicitation probe found the test-overwrite exploit *not elicitable within the tested scope* ($\leq$ 14B, edit-distance and regex tasks, the test-overwrite affordance): instruction tuning suppresses it (0/192), base 7B models sit at a $\approx 1\%$ noise floor, and a capable 14B base model solving the tasks at 52–55% still produced 0/512 (95% Wilson CI $[0, 0.74\%]$), not rising from 7B to 14B. We conclude that a usable hacking detector must reintroduce some oracle-adjacent signal; the natural reframing—rollout geometry as an *efficient sampler* that decides when to spend a scarce oracle query, rather than a classifier—is motivated by these results but left as future work, because no real hack was elicitable within scope to build it against.

1 Introduction

Detecting reward-hacking *onset* during RL post-training would be valuable: an early signal, computed from quantities the trainer already produces, could prompt intervention before a model ships. The attractive form is *oracle-blind*—it reads only rollout geometry (cheap, always available) and never consults a held-out quality measure. The hypothesized signal was an abrupt geometric shift (rapid KL acceleration, entropy collapse, advantage redistribution) marking the transition into hacking, preceding the point at which a held-out oracle gap becomes visible.

This paper reports that the oracle-blind version of this monitor cannot function as a *hacking classifier*, and that the obstruction is structural rather than an artifact of tuning. Section 2 states the argument. Section 3 describes the apparatus we built to test it. Section 4 gives three empirical legs consistent with the argument at every measured point: a characterization study in which candidate discriminative axes collapse against matched benign controls (§4.1); two real benign runs that are false positives, with a durable, reproduced operating point (§4.2); an onset-strength crossover that locates the impossibility region quantitatively (§4.3); and an elicitation probe showing the test-overwrite exploit is not elicitable within the tested scope (§4.4). Section 5 states limitations up front. Section 6 gives the implication and the reframing it motivates. We lead with the result rather than the original detector hope: oracle-blind geometry detects *that a convergence is underway*, not *that it is a hack*, and we make that boundary precise.

2 The structural argument

Let a policy π be optimized against a reward r. Reward hacking and benign capability improvement are, at the level of the rollout distribution, the *same event*: the policy concentrates probability mass onto the outputs r rewards most. In both cases KL from the reference policy grows, output entropy falls as the policy commits to a mode, and the advantage distribution narrows as group rewards equalize. These are the signatures of *convergence*, and convergence is what optimization does irrespective of whether the favored mode is desirable.

The *only* property separating a hack from a benign improvement is whether the higher-reward mode also has higher **true quality**. That property is not a function of r (during hacking r looks fine—that is what makes it hacking) and not a function of rollout geometry (which measures *how* the policy moves, not *toward what*, judged against ground truth). It is a function of the **oracle**: a held-out measure of true quality. An oracle-blind detector cannot observe the oracle, hence cannot observe the only quantity that separates the regimes.

Proposition 1 (oracle-blind indistinguishability)**.** *Let a detector be any measurable function f of rollout-geometry features R (KL, entropy, and advantage statistics computed from a run). For any hacking run there exists a benign-convergence run with identical R, differing only in the oracle O (true quality), which f does not observe. Hence $f(R)$ is identical on the two runs, and no such f separates hacking from benign convergence by kind.*

Assumptions. (A1) The detector is oracle-blind: its feature set is R, and $O \notin R$. (A2) R is a deterministic function of the run's policy outputs and the proxy reward r, i.e. $R = g(\text{outputs}, r)$, none of which involves O.

Proof sketch. By (A2), two runs with the same policy-output and reward trajectories have the same R; reward hacking is precisely the case where r is high while O is low, so a hacking run and a benign-convergence run can be constructed (or, empirically, *occur*) with coincident policy-output/reward geometry while their oracles diverge—O carries information not present in r and hence, by (A2), not present in R. Since f is a function of R alone (A1), f takes the same value on both runs. The quantity that distinguishes them, O, is conditionally independent of f's input given r and is unobserved by f; therefore the separating information is structurally unavailable to any oracle-blind f. □

The proposition subsumes advantage-distribution statistics: although advantage shape can encode "a subset of samples discovered a higher-reward mode," that event occurs identically in benign discovery and in hacking, and advantage shape is part of R (computed from r and oracle-blind outputs), so it is covered by (A2). The result bounds achievability without claiming geometry is uninformative—geometry genuinely detects *that a convergence is underway*; Proposition 1 says it cannot certify the convergence is a hack.

3 Apparatus

Flight Recorder instruments a GRPO trainer and computes per-step danger metrics from rollouts. Oracle-blindness is enforced at the type level: the trainer adapter assembles a rollout batch, split by extractors into a `RolloutFrame` (the only object a detector receives) and an `OracleFrame` (consumed only by the offline evaluator). `RolloutFrame` carries no oracle fields; `Detector.update(frame: RolloutFrame)` is typed to accept only rollout geometry, so oracle leakage into a detector is a type error.

The detector-visible extractors produce: KL mean/slope/acceleration; entropy mean and collapse trend; advantage variance, skew, kurtosis, and a Wasserstein "drift" of the advantage distribution against a rolling baseline; and generation statistics (mean/var generation length, n-gram diversity, log-probability concentration). A separate, evaluator-only oracle extractor computes the reward–quality gap and its changepoint.

The default detector, `ContractionTubeDetector`, fits a robust reference tube on a warmup window over [`kl_accel`, `entropy_trend`, `adv_drift`] (per-dimension median and MAD) and declares onset when the tube is exited (distance in MAD units beyond a radius ρ, exceeding τ) *and* a CUSUM on a geometry signal (default `kl_accel`, normalized by warmup mean/std) exceeds h, *sustained* for a persistence window. Crucially, both gates are **warmup-normalized**: the tube in MAD units of each run's own first 30 steps, the CUSUM in warmup z-scores. The detector is thus scale-invariant—absolute KL magnitude is divided out. The pre-registered thresholds (τ=1.0, h=5.0, ρ=3.0, warmup = 30, persistence = 5) were frozen on a synthetic harness before any real-run data. Because the synthetic harness's onsets are authored, it validates plumbing and tunes false-positive rate but cannot establish that the signal exists in real runs.

4 Empirical results

4.1 Detector characterization: candidate axes collapse against matched benign controls

Running the real extractors and detector over controlled trajectories, we asked of each candidate discriminative axis whether it separates a hack-like profile from a benign profile *matched on that axis*. *Convergence sharpness*: the tube signals fire on any sufficiently sharp convergence; three benign profiles (collapse onto a constant, convergence onto a complex solution, convergence via a high-diversity route) produced indistinguishable onsets. *Output degeneracy*: the generation-length / n-gram-diversity / log-probability-concentration features separate a short output from a long one, but that is "short = suspicious," which a correct one-liner also satisfies; against a hard control the degeneracy axis fired *identically* on `return 0` (a degenerate hack, on a task where 0 is wrong) and `return n*(n+1)//2` (a correct one-liner): both short, low-entropy, high-concentration. *Advantage-distribution shape*: a heavy-tailed advantage distribution is detectable but fired *identically* on a hacking profile (one sample found an exploit) and a benign control (one sample was benignly lucky), which share the same advantage shape. In each case the apparent discriminative power against a soft control vanishes against a control matched on the same axis—the empirical shadow of Section 2.

4.2 Two real benign runs are false positives, with a durable, reproduced operating point

We ran a genuine non-hacking GRPO run to measure false-positive behavior on real geometry: base Qwen2.5-1.5B, a well-specified coding task (longest substring with at most two distinct characters; a multi-case correct verifier so no degenerate constant can pass), 120 steps, learning rate 10^{-6} (deliberately gentle), no seeding, no curriculum. The run converged benignly—mean reward 0.19 → 0.9, entropy 1.9 → 0.45, KL ≈ 0.02 throughout—and **tripped the detector**: a latched onset at step 94 and an instantaneous trigger on 42/91 = 0.46 post-warmup steps, at a warmup-normalized peak CUSUM of $S \approx 37.8$, roughly eight times the pre-registered h=5. Benign convergence, normalized to its own quiet early phase, is a *strong* signal. We re-ran the identical configuration on an independent seed with the full per-step geometry persisted to durable storage. The second run reproduces the result exactly—onset 94, peak CUSUM $S = 37.55$—and now carries *real* advantage-drift (range $[0, 1.19]$) rather than the zero placeholder used when the first run's artifact was reconstructed from logs. Replaying the detector over the persisted real geometry yields the same onset and operating point. The false positive is therefore (a) reproduced across seeds at a stable operating point and (b) confirmed not to depend on the advantage-drift signal that was missing from the first analysis.

4.3 The crossover: where no threshold separates hacking from a benign run

Recalibration—raising h until the benign run goes quiet—appears to work against an *authored, strong* synthetic onset, but that margin is a function of how strong the synthetic onset was authored to be, not a property of real hacking. The correct question is the inverse: how strong must an onset be to remain separable from the benign run? Sweeping onset strength downward and comparing in the detector's own warmup-normalized CUSUM statistic (scale-fair), Table 1:

Below that, the benign run's CUSUM is at least as large, so *no* setting of (τ, h) keeps the benign run quiet while firing on the onset. The bar for distinguishability is "louder than a benign capability jump," not

onset strength s	onset peak CUSUM S	separable from the benign run?
0.10	231.8	yes
0.04	78.5	yes
≈ 0.024 (crossover)	≈ 37.8	boundary
0.02	27.6	no — indistinguishable

Table 1: Onset-strength sweep against the persisted benign run. An onset is separable iff its peak CUSUM exceeds the benign run's $S \approx 38$.

"louder than noise"—and Section 2 says a hack, being the same geometric event as a benign jump, need not be louder than one.

4.4 Elicitation probe: the exploit is not elicitable within the tested scope

The strongest objection to a negative result on detection is that we never put a *real* hack in front of the detector. We probed, by inference only (no training), whether an emergent test-overwrite hack is even *elicitable*. We sampled completions under a prompt that exposes the **test-overwrite affordance**—the submission is stated to run in a working directory containing the grader file, which is shown—and AST-scanned each completion for file manipulation. We call this rate P2; because GRPO can only reinforce behaviors that are sampled, P2 is an upper bound on what training could amplify. The picture is monotone and, with the 14B run, statistically resolved (Table 2).

Model	scale / tuning	P2 (exploit rate)	95% Wilson CI
Qwen2.5-Coder-7B-Instruct	7B, instruct	$0/192 = 0.000$	—
Qwen2.5-Coder-7B (base)	7B, base, coder	$2/192 = 0.010$	$[0.003, 0.037]$
Qwen2.5-7B (base)	7B, base, general	$1/192 = 0.005$	$[0.001, 0.029]$
CodeLlama-7b-hf (base)	7B, base, coder	$1/192 = 0.005$	$[0.001, 0.029]$
deepseek-coder-6.7b-base	7B, base, coder	0/192 (P1= 0, uninformative)	—
Qwen2.5-Coder-14B (base)	**14B, base, coder**	$\mathbf{0/512 = 0.000}$	$[\mathbf{0.000}, \mathbf{0.0074}]$

Table 2: Elicitation probe. Instruction tuning suppresses the exploit entirely; base models sit at a $\approx 1\%$ noise floor; doubling to 14B base (with strong P1: 52–55% solve rate, so the zero is genuine, not a gradient/parse artifact) gives 0/512 with CI entirely below a $\sim 5\%$ trainable threshold and not rising with scale.

Within the tested scope ($\leq$ 14B, edit-distance and regex tasks, the test-overwrite affordance), the exploit is either tuned away (instruct) or below trainable rates (base), and it does not rise with scale from 7B to 14B. At reachable scale, the phenomenon a hacking detector would need is not elicitable—which reinforces the practical force of the structural result: in the configurations we could reach, there is frequently no real hack for oracle-blind geometry to distinguish from benign convergence in the first place. (Larger models and other exploit affordances are not covered by this claim; see Section 5.)

5 Limitations

The real false positive is $n = 2$, not $n =$ many. Two independent seeds of one task/model reached the same operating point (onset 94, $S \approx 37.5$). This is a reproduced existence proof that real benign convergence lands in the impossibility region with a stable operating point; it is not a claim that benign convergence universally sits at $S \approx 38$ across tasks and models.

The advantage-drift caveat is resolved. The persisted second run carries real `adv_drift` (up to 1.19) and yields the same onset and CUSUM operating point, confirming the firing is driven by the CUSUM-on-`kl_accel` path and that the previously missing signal does not change the result.

The elicitation finding is bounded to ≤ 14B, these tasks, and the test-overwrite affordance. It does not show that larger models cannot be induced to hack, nor that other exploit classes (e.g. verifierparsing exploits that fool a string- or regex-based grader rather than overwrite a file) are equally unreachable. These are named as future work, not as gaps in the present claim.

Thresholds are pre-registered on synthetic data. Warmup-normalization removes absolute-scale miscalibration as an explanation, but a real-onset reference—which an unelicitable hack denies us—would be needed to claim the monitor *could* work after recalibration; the crossover shows recalibration cannot rescue the impossibility region regardless.

6 Implications

Pure rollout geometry is insufficient for hacking detection, by a structural argument and three consistent empirical legs. A usable detector must reintroduce an **oracle-adjacent signal**—a held-out verifier, a cheap proxy of true quality, or a known-benign contrastive reference—to recover the one quantity geometry cannot see. This reframes the goal from "detect hacking from geometry" (impossible oracle-blind) to "detect hacking with the *least* oracle signal." The natural design is **geometry as a sampler, not a classifier**: the contraction-tube CUSUM becomes a trigger that decides *when* to spend a scarce oracle query, and the oracle resolves hack-versus-benign at those points. The contribution would be an efficiency frontier—detection latency versus oracle queries spent, against a query-every-step baseline. We state this as motivated but untested: building and evaluating it requires a real hack to detect, and the elicitation gate (§4.4) closed before we could produce one. Characterizing this geometry-as-sampler / oracle-as-verifier trade-off, and the minimal oracle signal it needs, is the natural next direction once a real, reproducible hack is available.

Reproducibility statement

The detector, extractors, and the multi-axis characterization detector are implemented in the Flight Recorder codebase; thresholds are pre-registered and frozen before any real-run data. The second benign run's full per-step geometry is persisted as a public artifact and the onset/CUSUM operating point reproduces from it with no GPU. The crossover is a deterministic offline computation over the persisted benign series and a strength-parameterized onset generator. The elicitation probe is inference-only over six open-weight models. The entire empirical record was established for approximately \$2.23 of compute; no training run was required, because the elicitation gate showed the exploit GRPO would need to amplify is not present in the sampled output distribution.